

Implementation of Incident Management for Data Services Using ITIL V3 in Telecommunication Operator Company

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Abstract—The development of data services (internet) in Indonesia is increasing for each year. The research firm emarketer said that the population netter in Indonesia reached 83.7 million people in 2014, and in 2017 is estimated to reach 112 million people. pt. xyz is a company with business in telecommunications by providing data services as one of its business processes. But in that business process, often the emergence of an incident which resulted in delays in the services provided. Therefore this study was conducted to evaluate the incident management process applied by the company in conducting prevention / resolution of the incident. Incidents that use in this research is incident that cause interference in data service (internet) were received by users or result in failure of the system in the internal unit. ITIL incident management metrics supposed to be measured processes of incident management companies currently doing a resolution of an incident. itil is used as a guideline in the implementation and evaluation for lifecycle incident management process. the result is metrics kpi incident management is meet the target of resolution time but there are some points that need to be improved, such as the involvement of service desk that only 8% of the total incidents were resolved and the handling of incidents with the level of emergency and critical severity level.

Keywords— *incident management; ITIL; data services; incident*

I. INTRODUCTION

The development of Information Technology (IT) today is increasing not only in terms of software and hardware but also the data or Internet services. This is according to market research conducted by the research institute for professional digital business which eMarketer that says that the population of netizens in Indonesia reached 83.7 million people in 2014, and in 2017 is estimated to reach 112 million people. Even these figures beat the growing number of users in developed countries like Japan. The development of internet users in Indonesia is influenced by several factors such as the number of Indonesian population and also ease to access. By using mobile phones and mobile broadband connections are provided for the service provider enables users to use the Internet.

Seeing that a lot of companies - companies in telecommunications began to move quickly to doing imlementasi latest technologies to satisfy customers. One technology that is implemented today is 4G LTE, which allows

users to have Internet access speeds that reached 70 to 100 mbps at a certain frequency. The speed and stability of the Internet connection is one factor that is driving customers to continue using the service provided by the provider.

An IT-based services sometimes does not go in line with expectations. The existence of an unexpected case such an incident would hinder processes that are running. incident is a condition where the incidence of errors that are not in accordance with the conditions expected, as was the case of system failure, no operation of a process, or the discovery of the problem [1] incidents are usually the result of a system failure or error on the IT infrastructure is causing or potentially cause operational disruption.

Common problems frequently encountered is the lack of attention in the handling of an incident into account. Handling incidents of poor management will lead to the parties assume that the investments made in the IT side is less positive impact [2]. Disruption of service received by customers due to the occurrence of an incident will form a bad opinion if the incident cannot be resolved quickly.

It is necessary for the incident management process as a reactive step in preventing and combating the event of an incident. According to [3] reactive management can be seen as two major events in the control comprising:

Identification - The management of the received information on incidents reported from the incident management process and service desk. Problem management team members analyze this information, look for similarities in the symptoms reported incidents. They are looking for a record previously identified problems that could explain the symptoms. If none can be found, a note explaining the problem is newly created.

Classification - control activities have identified the importance of new issues and designate resources to handle them.

According to [4] Incident Management consists of four main parts, namely an emergency call, tell the one who is responsible, do response and repair.

Incident management usually closely related to Service Desk, which is the point of contact for all IT users to

communicate with. When service is interrupted or failed to provide the promised performance during normal service hours. Service desk can investigate and analyze all information related to the incident [5]. It's important to restore services to normal operation as quickly as possible.

There are some standards that can be used as a reference in the management process incident. One standard that is quite popular is the Information Technology Infrastructure Library (ITIL). ITIL is a framework that provides a practical step in the governance of IT, and perform control in an IT service. The definition according to [6] Information Technology Infrastructure Library (ITIL) is a framework or a framework that provides a set of models of best practice processes and functions of IT service delivery.

XYZ is one provider of mobile telecommunications operators are now seen as one of the mobile service provider for Internet and Telephone in Indonesia. As a step in providing good quality service, implement incident management process as a basic step in maintaining the quality of services provided. This study will refer to the Incident Management process which has been carried out and is expected to be addressed by improving the quality of the Incident management that has been running at the moment.

II. LITERATURE REVIEW

A. ITIL Framework

ITIL (Information Technology Infrastructure Library) is a framework that provides practical steps on the activity of IT Service Management. In a recent survey of 491 members of the IT Service Management Forum (itSMF), it was found that ITIL favorable productivity improvement services (85%), quality (83%), business reputation (65%) and cost savings (41%) [6]. Framework this provides a step in terms of governance, management, and control of iT, with a focus on the development and improvement of a quality of iT services to be provided [7].

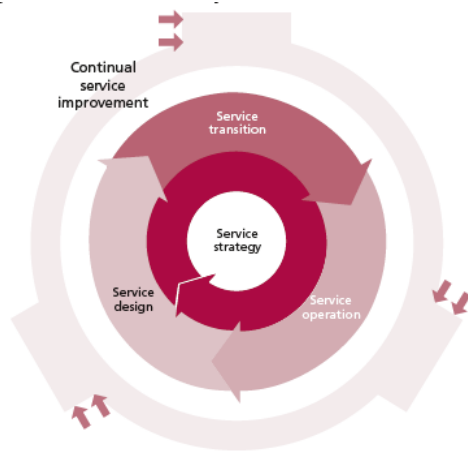


Fig. 1. ITIL V3 Service Lifecycle

B. Incident Management in ITIL V3

Activities in incident management (incident management) according to the ITIL v3 framework as also shown in the figure below) [9].

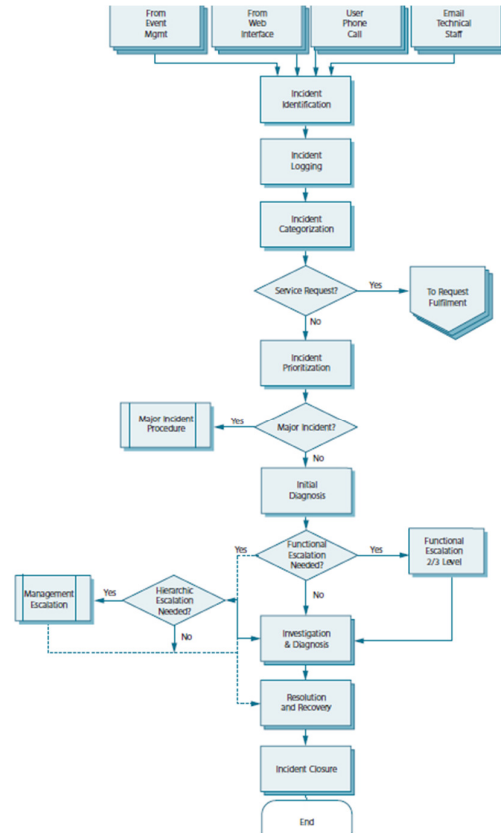


Fig. 2. Incident Management Process flow

The following is an explanation of each - each step in the Incident Management Process flow:

- Identification of the incident (incident identification)
The process of incident management begins with identifying through technical employees or service desk. Moreover field identification can be obtained by looking at the alarm originating from network devices. It should be noted whether the alarm was appropriate and in accordance with the incident that occurred on the field.
- Recording of the incident (incident Logging)
Recording is done to see if the incident is most often encountered both large and small incidents. Some of the information that should be noted is related to an incident ID, incident category, time happens, incident description, name of the person / group responsible for the handling, the implications of the incident, and the time of closing the case.
- Categorizing incidents

See how big an impact that can result from an incident and are categorized with the aim of prioritizing ease in handling. Category incidence can be made based on the estimated duration of treatment, the implications for the organization's business processes, and the number of technical staff related.

- Priority incidents

Determine the priority of an incident based on the categorization in the previous step. Acceptable impact for the customer or long treatment should be done can be prioritized in different categories depending on the related divisions

- Early diagnosis

Early diagnosis is basically done by the front office where the division that receives the customer complaint and attempt to resolve it if it can be done. If the incident is found by telephone from the user, then attempted to resolve the incident during a phone user still in touch.

- Escalation incident

If the front office failed to resolve the incident so they can raise the level of incident handlers to be escalated by the technical side. Escalation of the incident consists of two kinds of escalation and escalation functions hierarchy. Escalation function is to raise the level of escalation into a top level, in contrast with the escalation hierarchy across the organizational hierarchy eg directly to the IT manager or managers associated with the incident.

- Resolution (resolution and recovery)

These actions are the actions taken to resolve an incident. Recovery steps taken parties associated in any incident, such as the technical staff who carry out the network configuration or device supplier if an error occurs on the device used.

- Closure (incident closure)

Closing step is a step undertaken by the service desk and technical staff linked to ascertain whether the incident had been handled properly completed. Which must be considered in the closing strides this is the documentation of the process of handling the incident, an estimate of the recurrence of incidents, and user satisfaction survey on the handling of the incident.

III. RESULT AND DISCUSSION

The main goal of the Incident Management process is to restore normal service operations as quickly as possible and minimize the adverse impact on business operations, thus ensuring that it is likely the best level of service quality and availability is maintained.

It is important for stakeholders to provide consistent processes to track incidents and ensure that:

- Incidents have been recorded
- incident response directed to the right person

- Incident Status accurately reported
- Unresolved incidents can be viewed and reported
- Incident correctly prioritized and handled in the proper order

In relation to the manufacture of table priority it places a high incidence of clear definitions in the following

1) Customer

The customer or the customer is a person who use and purchase of services provided by the company. In this case the customer consists of a VIP or the council rideksi, public as mobile data users, and the company also formed a partnership.

2) Impact or impact

Impacts are divided over how many personnel or impaired function will be incidents. Can be divided into three categories, namely:

- Low: One or two personnel where the services were experiencing disruptions but still operates in accordance with the Service Level Agreement (SLA) established
- Medium: Many personnel in one location simultaneously. The function of the service is still running but the disorder that results of operations performed is not in accordance with the specified SLA.
- High: All users in a particular service. Personnel on some services disrupted. And the unavailability of the services provided.

3) Incident Category

An incident is a disorder that is not planned for an IT Service or a reduction in IT Service Quality. The failure of any item, software or hardware, which is used in support of a system that had not yet affected service is also an incident. There is table of category for each trouble ticket.

Category	Sub Category	Total TT
EnodeB Down	Alarm EnodeB to SGSN down	46
	EnodeB to MME down	8
NE Unmonitor	RNC Unmonitored	5
	EnodeB Unmonitored	10
	SGSN Unmonitored	7
	BSC Unmonitored	8
Customer Complaint	Connection to Internet	10
	Network Priority	10
	Network Profile	14
	APN Network	4
	Alarm GbIP	166

Link MPLS (Transport)	Alarm IuPS	74
	Traffic Upstream	27
Trouble in DNS	DNS Link Down	3
Service Quality	Service Degradation	9
Total		401

4) Incident Repository

Incident Repository is a database that contains relevant information about the incident that has diselesai or unfinished tackled.

5) Priority

Priority is given to an incident that will determine how fast the response time to a resolution that depends on the combination of the severity of the incident and the resulting impact.

Network Unit	Site Priority	Description	
BTS / Node B	P1	High revenue or Hub (10 or greater) Sites	
	P2	Medium Traffic	Medium revenue or HUB (Less than 10) Sites
	P3	Low traffic	Regular sites, end-sites which is able to define as LOW and medium revenue sites.
BSC / RNC	P1	High traffic	Urban
MSC / HLR / IN / IGW	P1	High, Medium traffic	
SGSN / GGSN	P1	High, Medium traffic	
Transmission links - Microwave	P1	High traffic	Equal or > 10 BTS
	P2	Low traffic	Less than 10 BTS
Transmission links - Optical	P1	All traffic	

6) Response and Resolution

A response time required of an incident to be reported and given to a person or the division who are capable of doing prevention.

While the Resolution is an activity undertaken not only in enabling customers to do their job according to the services provided but also steps to find the source of the problem from an incident to be identified and a solution found.

7) Severity

Severity is divided by the number of customers who are disturbed in their work. There are three categories in determining Severity namely:

- Low: issues that resulted in the customer fails to do some of the work he does.
- Medium: issues that resulted in the customer fails to perform the function - a function that is critical. As a system failure or disruption of business processes in a function.
- High: The main services provided are not

ITIL Incident Management Metrics

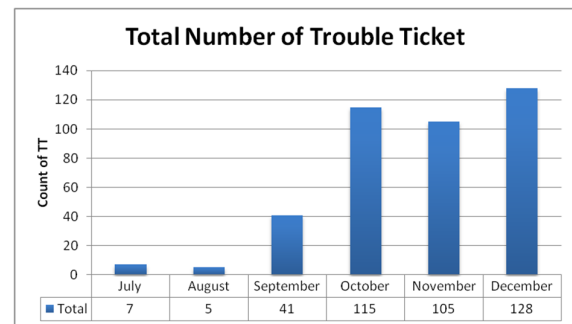
Incident management metrics used to review the Process of incident management in several factors such as efficiency to resolve the incident and effectiveness to response the incident. Total of 401 Trouble Ticket in period July - December 2016 will be validate to find the solution to improve process in incident management. That is the table for analysis component of KPI in that company:

KPI Name: Total Number of Incident

Definition: Total of incident completed as a control measure

Formula: All total incident per Monthly

Result:



KPI Name: Percentage of Trouble handled within target resolution time

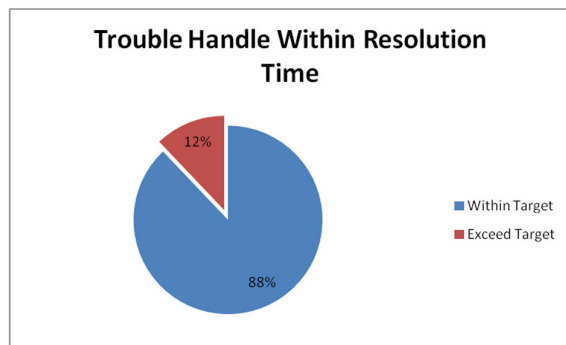
Definition: The percentage of troubles handled within target response time.

Formula:

$$SuccRate = \frac{RestoredIn\ Time(t)}{Total\ \#\ Of\ TT(n)} \times 100\%$$

- TT(n) Number of trouble tickets which are created in given time period
- RestoredIn Time(t) : Number of TT identified in given time period and be closed within target time.
- Measure : Percentage

Result:



Category	Summary	
	Within Target	Exceed Target
EnodeB Down	42	12
NE Unmonitor	29	1
Customer Complaint	38	0
Link MPLS (Transport)	222	31
DNS Trouble	1	3
Service Quality	9	0
Total	341	47

KPI Name: Average duration of trouble tickets by priority

Definition: The average duration from creation to closure for the trouble tickets of each priority.

Formula:

$$AverageDur(t, Priority) = \frac{\sum Dur(TT(t, Priority))}{Total\#\ OFTT(t, Priority)}$$

- TT (t, Priority): Trouble which priority matches Priority and submitted in given time period.
- Dur (TT(t, Priority)): the duration from Trouble submitted until closed or rejected.
- Priority: Critical, High, Medium, Low, etc. This priority list will be defined during level 5 regarding different requirements or policy in different site

Result:

- EnodeB Down

Category	P1 – Major TT		
	Average Time (HH:MM)	Priority Time	Result

Alarm EnodeB to SGSN down	3:03	3:00	1.01
EnodeB to MME down	2:53	3:00	0.96

- NE Unmonitor

Category	P1 – Major TT		
	Average Time (HH:MM)	Priority Time	Result
RNC Unmonitored	2:17	3:00	0.78
EnodeB Unmonitored	0:00	0:00	0
SGSN Unmonitored	1:05	3:00	0.53
BSC Unmonitored	0:14	3:00	0.51

- Customer Complaint

Category	P1 – Major TT		
	Average Time (HH:MM)	Priority Time	Result
Cant connect to Internet	1:22	8:00	0.82
Network Priority	1:19	8:00	0.16
Profile Network	1:49	8:00	0.22
APN Network	1:15	8:00	0.33

- Link MPLS (Transport)

Category	P1 – Major TT		
	Average Time (HH:MM)	Priority Time	Result
Alarm GbIP	2:43	3:00	0.91
Alarm IuPS	2:19	3:00	0.77
Traffic Upstream	1:57	3:00	0.65

- Trouble in DNS

Category	P1 – Critical TT		
	Average Time (HH:MM)	Priority Time	Result
DNS Link to SGSN Down	2:13	1:00	2.21

- Service Quality

Category	P1 – Major TT		
	Average Time (HH:MM)	Priority Time	Result
Service Degradation	1:11	3:00	0.6

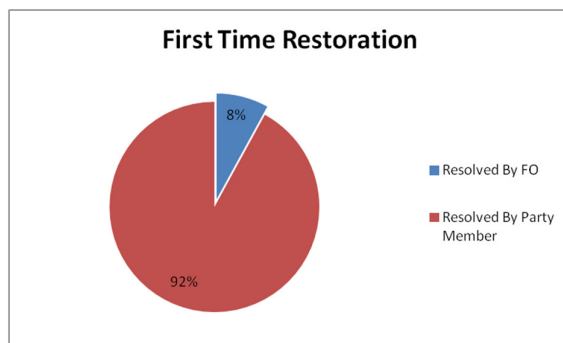
KPI Name: First time restoration

Definition: Number of TTs that are restored first time by FO after opening Formula:

$$FTR = \frac{\text{\# of TTs Restoration By FO}}{\text{Total\# OTT}}$$

- FTR: First time restoration
- Total # Of TT: Total Number of TTs handled

Result:



IV. CONCLUSION

Based on the analysis of data and the preceding discussion, the research findings can be summarized as follows:

- Processes that could be improved in the incident management at PT.XYZ is the initial stages of diagnosis and investigation. Solutions that can be offered is to implementations KEDB (Known Error Database) as one of the steps that can be taken to improve upon a good understanding about incidents and facilitate in identifying

incidents. And make a special treatment that can be implemented quickly to the incident with severity level Emergency and Critical order handling well managed. The need to make sure that every alarm on each network device is functioning properly in giving notification when interference with the devices.

- The incident that has the potential for great impact in the disruption of the company's business processes are incident with category Link MPLS (Transport). This is because these categories have a total number of incidents that most large compared to other categories, or nearly half of the total incidents from this category. If this category can be managed properly, it can lower the total of the existing trouble ticket and enhance the company's achievement of SLA targets. Management of incidents can be further developed with the identification of risks to facilitate the acknowledge
- From ITIL Incident Management Metrics show that the service desk as the first level in the process of incident management should be improved in terms of the involvement of a resolution of an incident. It can be seen from the value of involvement in First time restoration that only 8% of the entire trouble ticket. On handling the incident with severity level of emergency and critical also needed special attention, because of the trouble ticket with the severity level exceeds the target resolution specified.

REFERENCES

- [1] G. Eason, B. Noble, and I.N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529-551, April 1955. (references)
- [2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68-73.
- [3] I.S. Jacobs and C.P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G.T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271-350.
- [4] K. Elissa, "Title of paper if known," unpublished.
- [5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740-741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].
- [7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.